DSCI 552 Homework 5

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AI Usage Disclosure

I used AI tools to help me learn the relevant programming techniques and machine learning concepts, understand code implementations, and assist with debugging.

AI also assisted in constructing, executing, checking, and documenting this submission. All reported numerical results are produced by the executed code below. The supplied homework PDF does not prescribe an additional disclosure format.

Requirements and experimental protocol

The two-page assignment and all five footnotes were read, as were both supplied .names files and the supplied KDnuggets article. Local data are used without downloading replacements. The original question order is retained: 1(a) data, 1(b) full tree, 1(c) all rules, 1(d) pruning; 2(a) ordered split, 2(b) imputation, 2(c) correlations, 2(d) CV, 2(e) 11 feature plots, 2(f) OLS, 2(g) Ridge, 2(h) both LASSOs, 2(i) PCR, 2(j) XGBoost.

Seed 552 is fixed. All regression models retain all 122 predictive variables as candidate inputs; the 11-variable CV ranking is descriptive only. Regression tuning uses the same shuffled five-fold KFold within the training portion, minimizing validation MSE. The mandated external split is never shuffled. Imputation, scaling and PCA are fitted separately in each training fold through Pipeline. The test set is evaluated once per final fitted model, after all six models are frozen. No test metric is used to revise a model.

Dependency	Version
Python	3.13.5
numpy	2.5.4
pandas	3.1.0
scipy	1.18.1
scikit-learn	1.9.1
xgboost	3.4.1
matplotlib	3.11.2
seaborn	0.13.2
nbformat	5.11.1
nbclient	0.11.0
ipykernel	7.4.0
reportlab	5.1.0
pdfplumber	0.11.10
pypdf	6.21.4
pypdfium2	5.14.0
threadpoolctl	3.7.0

1(a) Acute Inflammations: data and label powerset

The actual byte encoding is detected from the BOM (UTF-16 if present, otherwise UTF-8); this takes precedence over the ASCII statement in diagnosis.names. Tabs separate fields. Decimal commas in temperature are converted to decimal points. The five binary symptoms and two diagnoses map no to 0 and yes to 1. The label powerset code is 2*d1+d2, so both original diagnoses remain identifiable. d1 denotes inflammation of urinary bladder; d2 denotes nephritis of renal pelvis origin. These are expert-created illustrative records, not evidence of clinical performance.

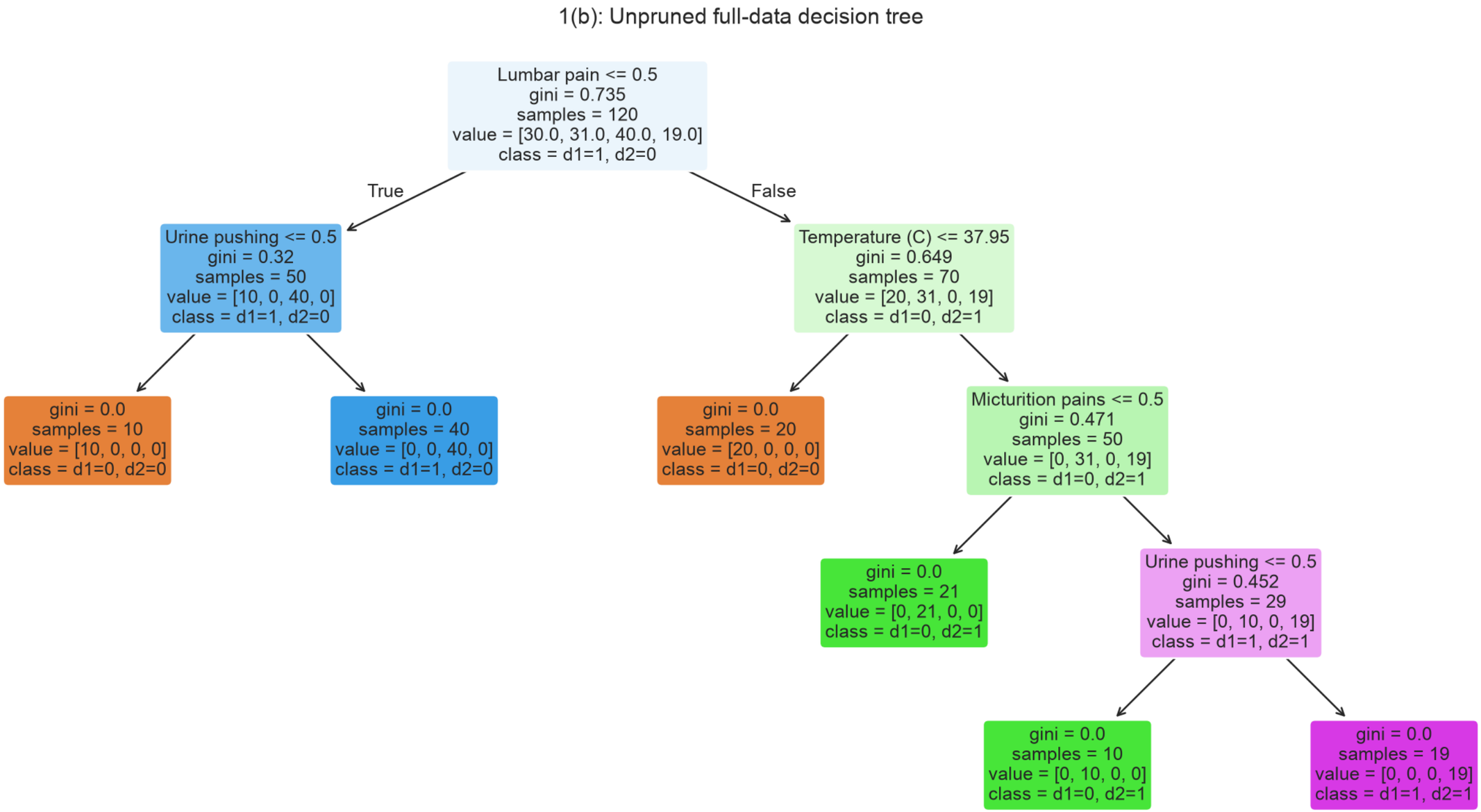
Powerset class	Bladder inflammation (d1)	Renal pelvis nephritis (d2)	Records
0	no	no	30
1	no	yes	31
2	yes	no	40
3	yes	yes	19

Validated: 120 rows, 6 predictors, 2 labels; encoding **utf-16**; no missing values.

1(b) Full-data decision tree

A CART classifier uses Gini impurity and the powerset target, with ccp_alpha=0 and otherwise unrestricted growth. As requested, the final tree is fitted to all 120 records. Tree class labels explicitly show both diagnoses (0=no, 1=yes). Training accuracy is a description of fit, not a held-out estimate.

1(b): Unpruned full-data decision tree All nodes and both diagnosis labels are shown.



1(c) All root-to-leaf IF-THEN rules

Each conjunction is one complete path. Binary branches are translated back to yes/no, and leaf classes back to the original two diagnoses. Temperature thresholds retain the model threshold to sufficient precision. n is the training support of the leaf. The executable path conditions are checked against tree predictions for every record, with exactly one matching rule per record.

Rule 1. IF Lumbar pain = no AND Urine pushing = no THEN bladder inflammation = no; renal pelvis nephritis = no (n=10).

Rule 2. IF Lumbar pain = no AND Urine pushing = yes THEN bladder inflammation = yes; renal pelvis nephritis = no (n=40).

Rule 3. IF Lumbar pain = yes AND Temperature (C) <= 37.950001 THEN bladder inflammation = no; renal pelvis nephritis = no (n=20).

Rule 4. IF Lumbar pain = yes AND Temperature (C) > 37.950001 AND Micturition pains = no THEN bladder inflammation = no; renal pelvis nephritis = yes (n=21).

Rule 5. IF Lumbar pain = yes AND Temperature (C) > 37.950001 AND Micturition pains = yes AND Urine pushing = no THEN bladder inflammation = no; renal pelvis nephritis = yes (n=10).

Rule 6. IF Lumbar pain = yes AND Temperature (C) > 37.950001 AND Micturition pains = yes AND Urine pushing = yes THEN bladder inflammation = yes; renal pelvis nephritis = yes (n=19).

1(d) Cost-complexity pruning and interpretable minimum

CART minimizes training leaf impurity plus alpha times the number of leaves. To define a useful minimum, a predeclared grid of 401 alpha values from 0 through 0.4 is evaluated by shuffled stratified five-fold CV on exact-match powerset accuracy. This grid is independent of label-derived pruning breakpoints. Select the largest alpha whose mean accuracy is within one standard error of the best mean; ties therefore favor fewer leaves. The standard error is the sample standard deviation of five fold accuracies divided by sqrt(5). This is a practical simplicity rule, not a proof of globally minimal trees or statistical equivalence. Final candidates are then refitted on all 120 records. The full-data pruning path is reported descriptively, not used to construct the CV grid. A stump is explicitly compared to the chosen model.

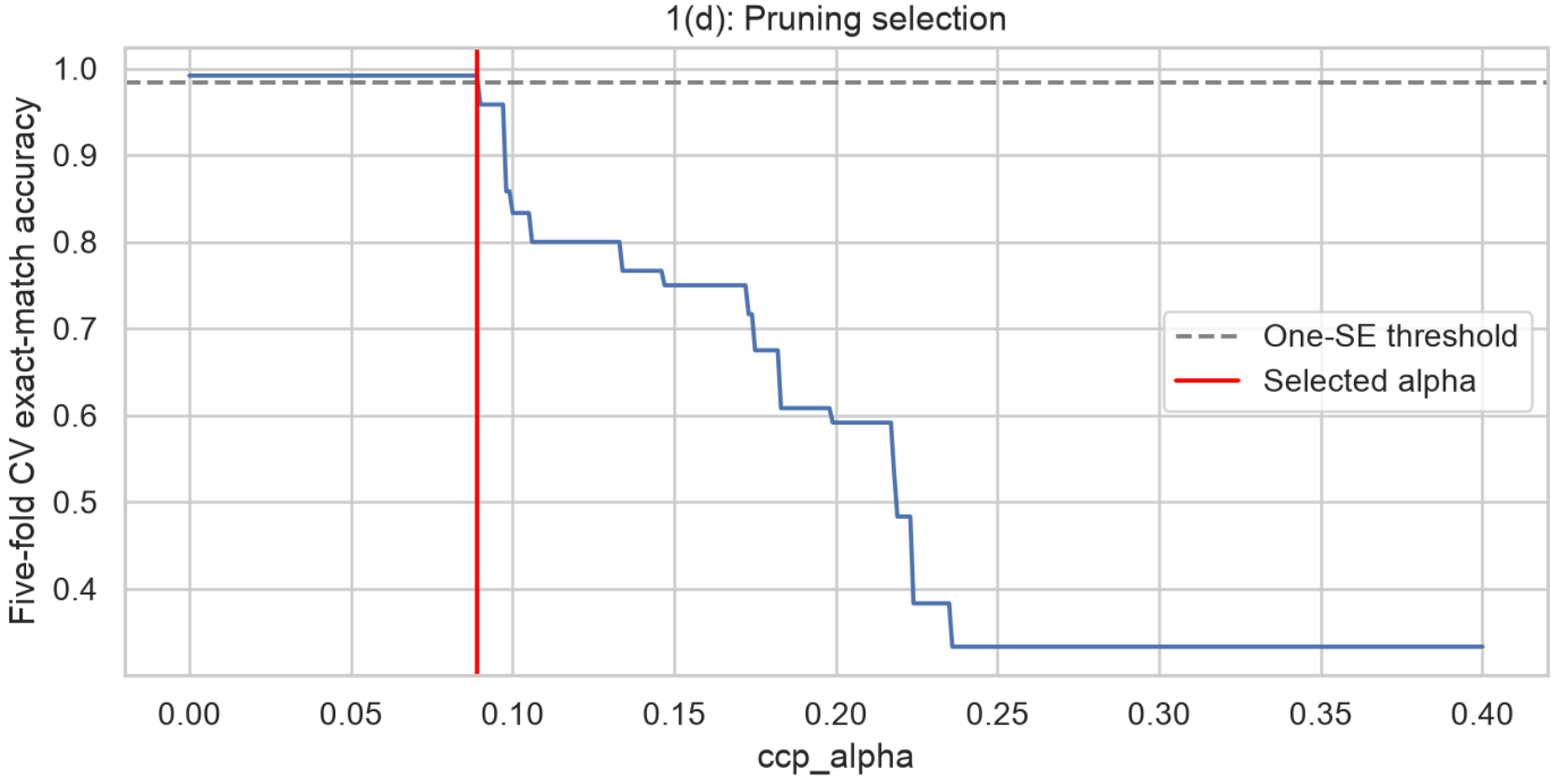
alpha breakpoint	Leaf impurity	Nodes	Leaves	Depth	Train exact accuracy
0	0	11	6	4	1
0.0981667	0.196333	7	4	2	0.841667
0.133333	0.329667	5	3	2	0.758333
0.182	0.511667	3	2	1	0.591667
0.222917	0.734583	1	1	0	0.333333

Tree	alpha	Nodes	Leaves	Depth	Train accuracy	Train Hamming loss	CV accuracy
Unpruned	0	11	6	4	1	0	0.991667
Selected	0.089	11	6	4	1	0	0.991667
Stump baseline	1	1	1	0	0.333333	0.4625	0.333333

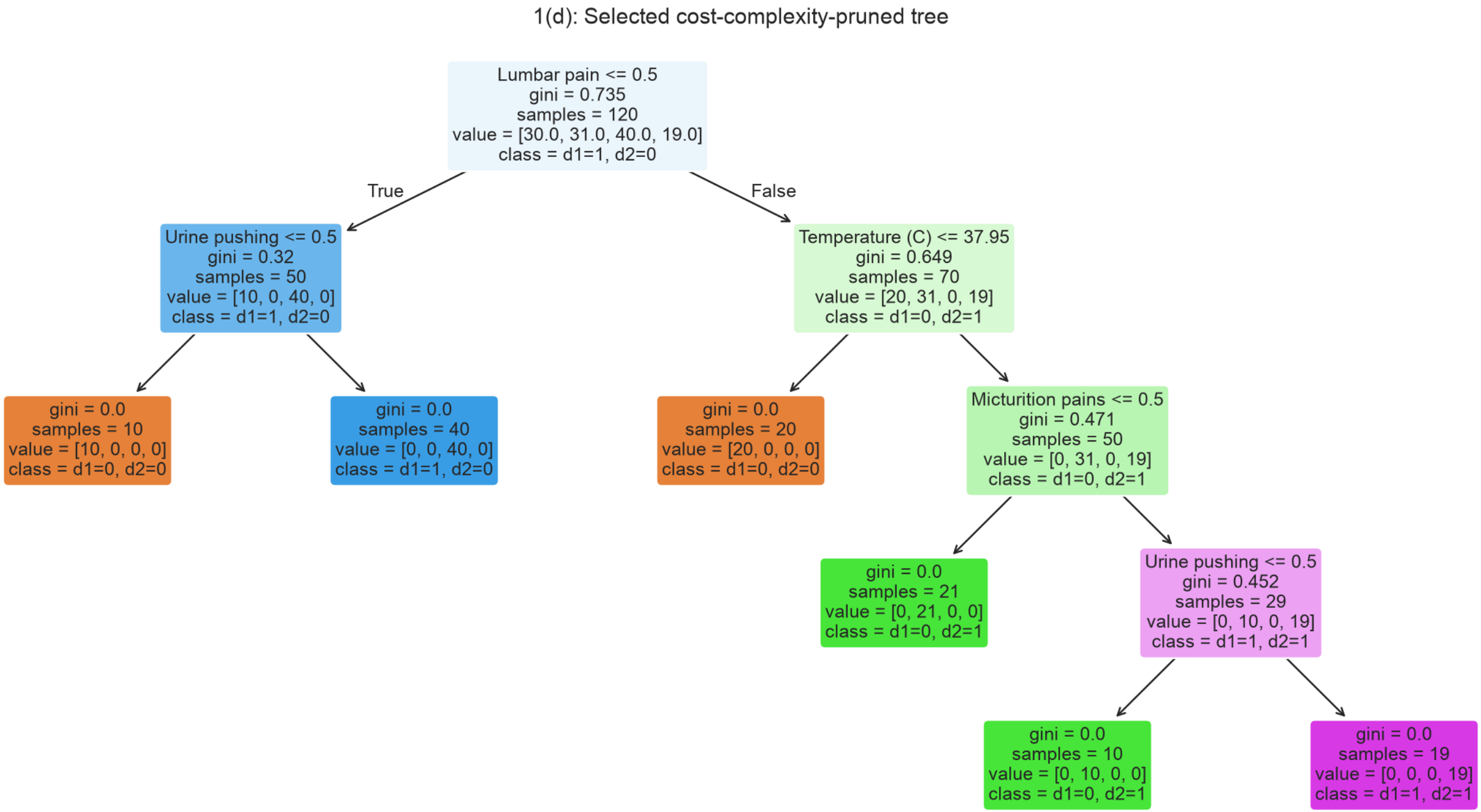
The best CV accuracy is **0.9917**, SE **0.0083**; the acceptance threshold is **0.9833**. Selected alpha = **0.089**. The selected tree has **6 leaves** and depth **4** versus **6 leaves** and depth **4** before pruning. No smaller subtree meets this criterion; keeping the original structure is preferable to sacrificing useful accuracy.

These CV scores informed alpha selection and are not an independent post-selection performance estimate. There is no separate test set in Question 1, in accordance with the full-data fitting requirement.

Alpha selection using only the acute-inflammations data.



1(d): Selected cost-complexity-pruned tree All nodes and both diagnosis labels are shown.



Pruned rule 1. IF Lumbar pain = no AND Urine pushing = no THEN bladder inflammation = no; renal pelvis nephritis = no (n=10).

Pruned rule 2. IF Lumbar pain = no AND Urine pushing = yes THEN bladder inflammation = yes; renal pelvis nephritis = no (n=40).

Pruned rule 3. IF Lumbar pain = yes AND Temperature (C) <= 37.950001 THEN bladder inflammation = no; renal pelvis nephritis = no (n=20).

Pruned rule 4. IF Lumbar pain = yes AND Temperature (C) > 37.950001 AND Micturition pains = no THEN bladder inflammation = no; renal pelvis nephritis = yes (n=21).

Pruned rule 5. IF Lumbar pain = yes AND Temperature (C) > 37.950001 AND Micturition pains = yes AND Urine pushing = no THEN bladder inflammation = no; renal pelvis nephritis = yes (n=10).

Pruned rule 6. IF Lumbar pain = yes AND Temperature (C) > 37.950001 AND Micturition pains = yes AND Urine pushing = yes THEN bladder inflammation = yes; renal pelvis nephritis = yes (n=19).

2(a) Communities and Crime: data and ordered split

Read the 128 attribute names directly from @attribute lines in communities.names. Keep original file order: training is original rows 1-1495, test is rows 1496-1994. The target is ViolentCrimesPerPop. The provided data were already normalized approximately to [0,1]; this is different from standardization to zero mean and unit variance. The data combine the 1990 Census, 1990 LEMAS and 1995 FBI UCR; normalization parameters used by the original dataset creator cannot be undone. All additional preprocessing here is learned solely from training data.

Validated raw shape: **(1994, 128)**. Training: **(1495, 122)**. Test: **(499, 122)**. Target excluded from predictors; all five nonpredictive fields removed.

2(b) Missing values and imputation

Question marks are parsed as NaN. Median imputation is robust to skew and retains all 122 predictors, including sparsely observed LEMAS variables. A median cannot reconstruct missing information or remove selection bias: LEMAS coverage is highly nonrandom according to the supplied documentation. No rows or predictive columns are dropped. An all-missing column within a training fold is retained and filled with zero via keep_empty_features=True (the audit below checks whether this fallback is actually needed). The descriptive training matrix is median-imputed using training rows only. Predictive CV uses a separate SimpleImputer inside every Pipeline; the descriptive matrix is never passed to GridSearchCV.

Feature	Train missing	Train missing fraction
OtherPerCap	1	6.689e-04
LemasSwornFT	1256	0.840134
LemasSwFTPerPop	1256	0.840134
LemasSwFTFieldOps	1256	0.840134
LemasSwFTFieldPerPop	1256	0.840134
LemasTotalReq	1256	0.840134
LemasTotReqPerPop	1256	0.840134
PolicReqPerOffic	1256	0.840134
PolicPerPop	1256	0.840134
RacialMatchCommPol	1256	0.840134
PctPolicWhite	1256	0.840134
PctPolicBlack	1256	0.840134
PctPolicHisp	1256	0.840134
PctPolicAsian	1256	0.840134
PctPolicMinor	1256	0.840134
OfficAssgnDrugUnits	1256	0.840134
NumKindsDrugsSeiz	1256	0.840134
PolicAveOTWorked	1256	0.840134
PolicCars	1256	0.840134
PolicOperBudg	1256	0.840134
LemasPctPolicOnPatr	1256	0.840134
LemasGangUnitDeploy	1256	0.840134
PolicBudgPerPop	1256	0.840134

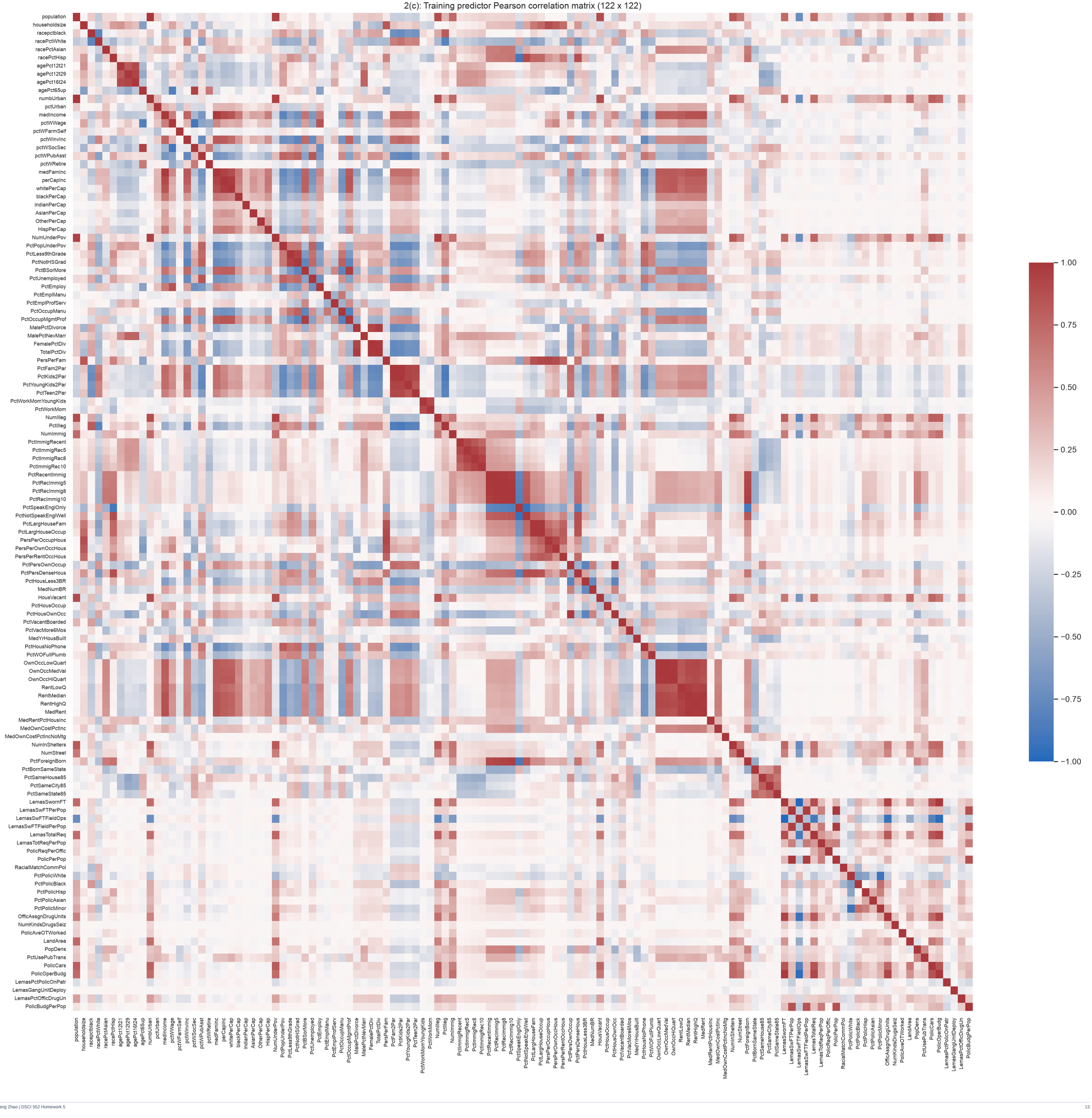
All-missing training-fold columns across five folds: **0**. Number of predictive columns dropped: **0**.

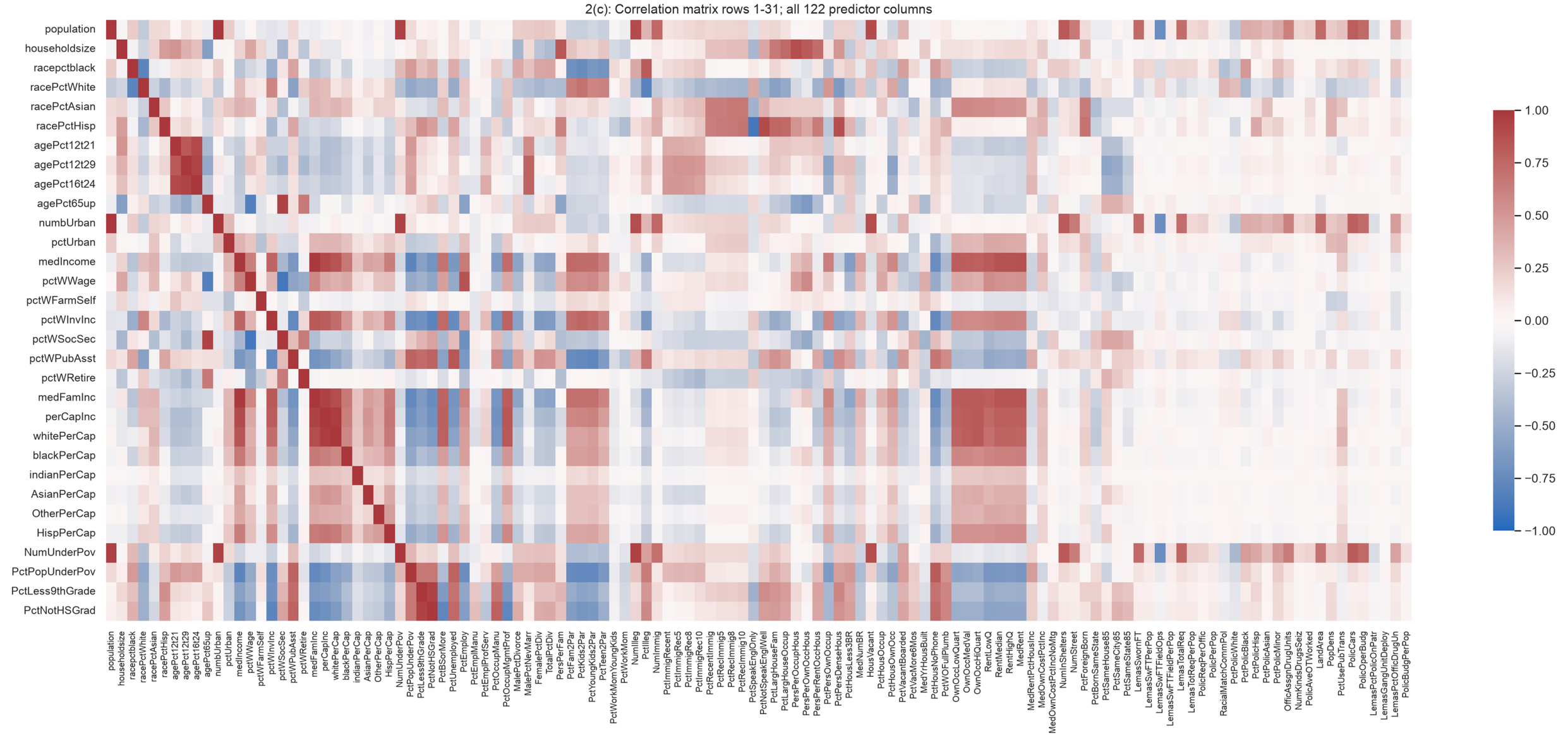
2(c) Feature correlation matrix

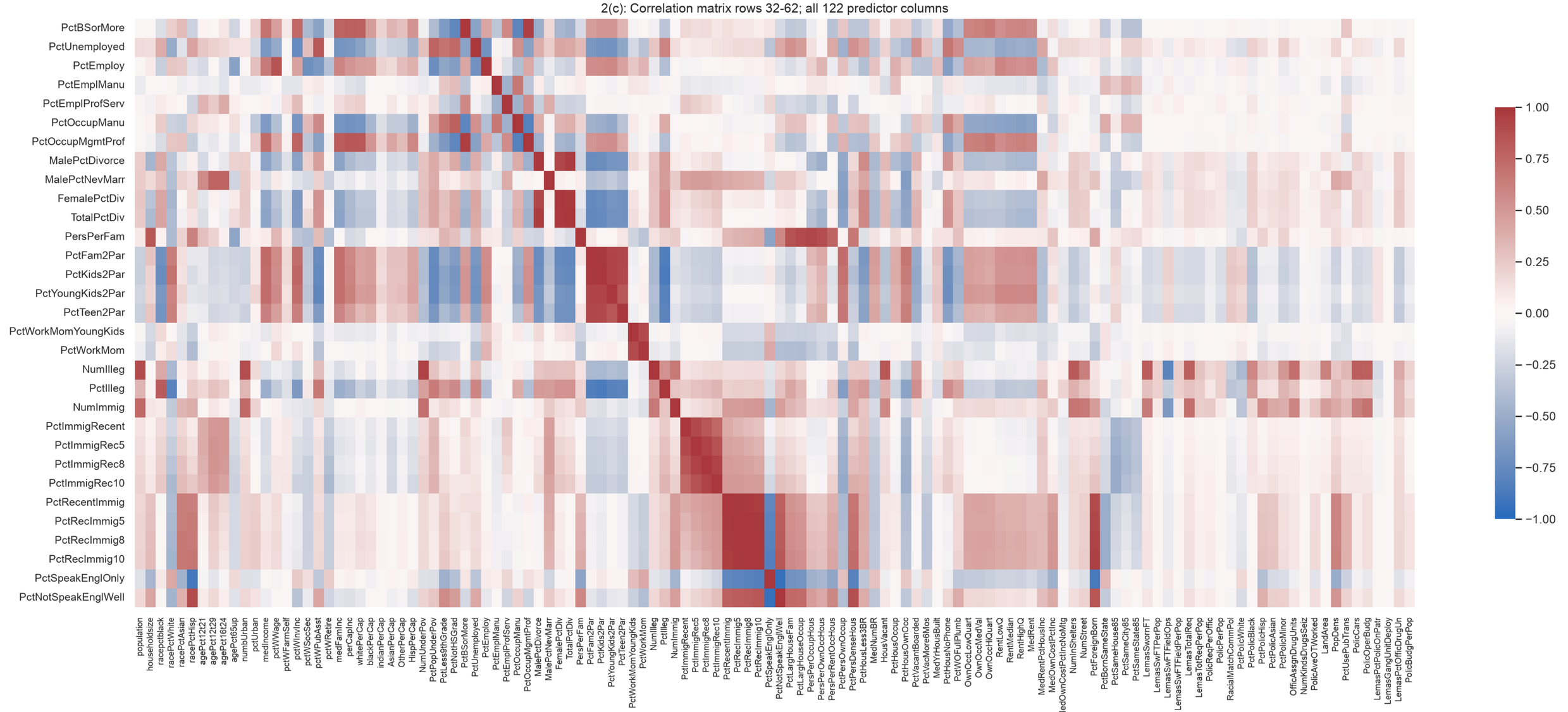
Pearson correlations below use the median-imputed 1495-row training matrix, before standardization. The target is excluded. All 122 feature labels are shown in a large overview and four enlarged row panels; the complete numeric matrix is also exported. Strong blocks of correlation indicate redundant predictors and motivate regularization and PCR. Imputation can alter correlations, especially for sparse LEMAS fields. These correlations describe association, not causation.

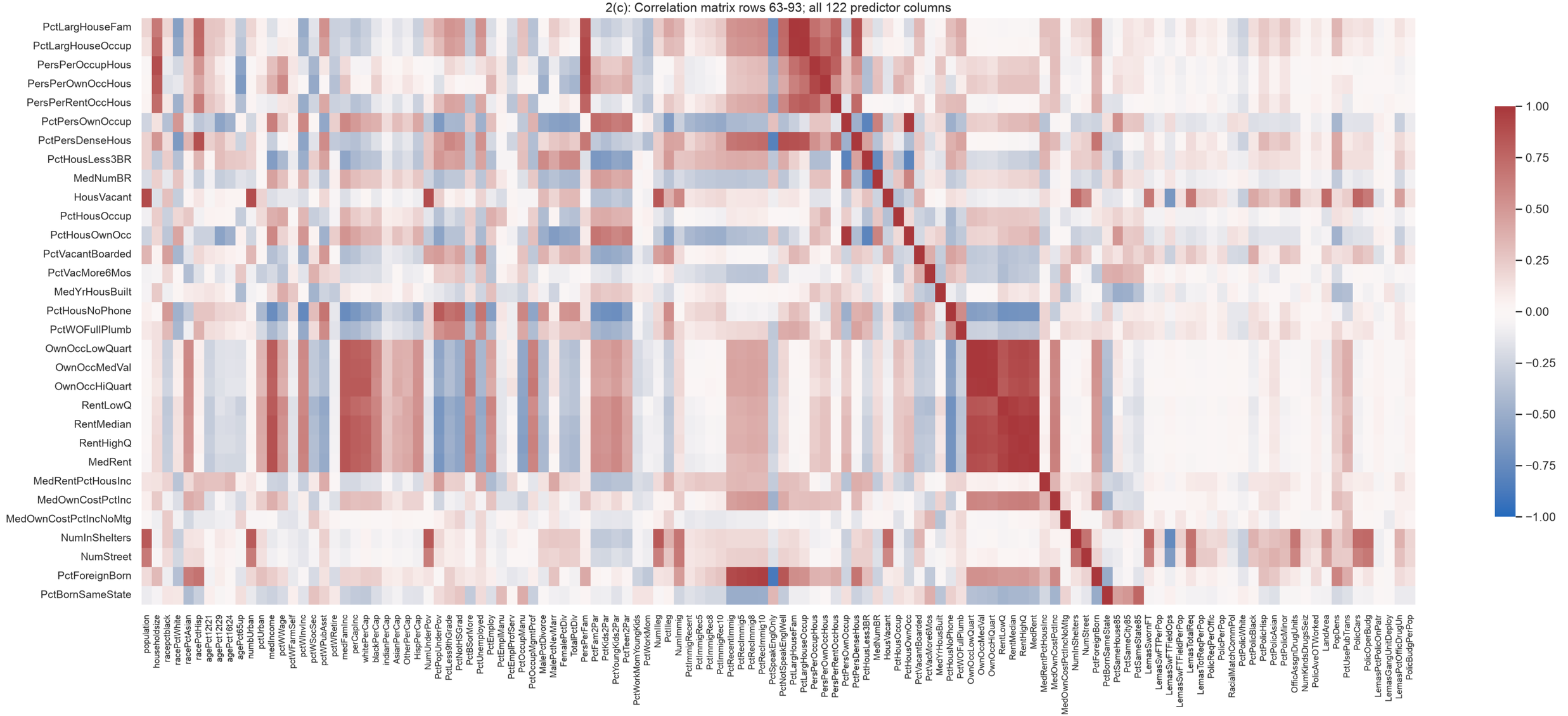
Full training feature correlation matrix; use the following enlarged panels for labels.

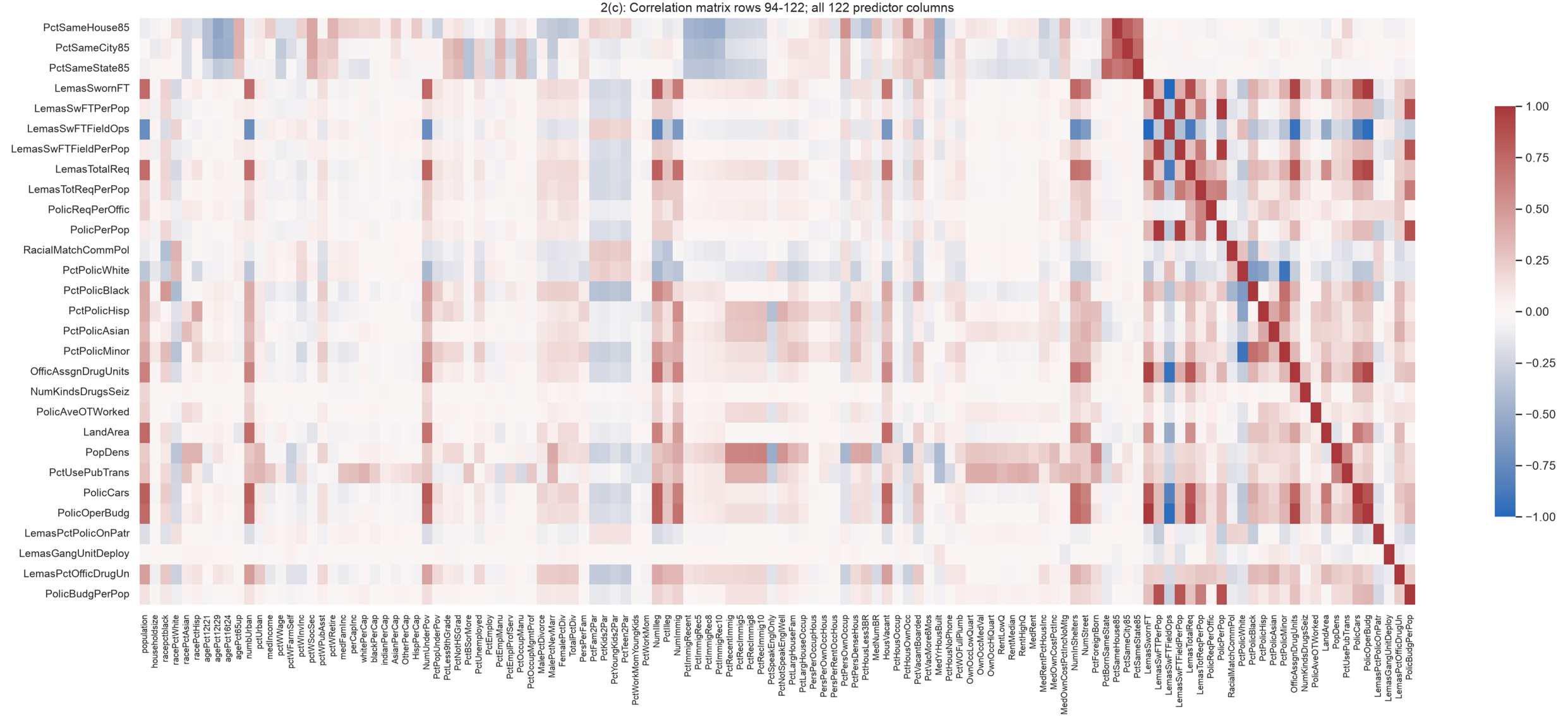
2(c): Training predictor Pearson correlation matrix (122 x 122)




2(c): Correlation matrix rows 1-31; all 122 predictor columns
population
householdsize
racepctblack
racePctWhite
racePctAsian
racePctHisp
agePct12t21
agePct12t29
agePct16t24
agePct65up
numbUrban
pctUrban
medIncome
pctWWage
pctWFarmSelf
pctWInvInc
pctWSocSec
pctWPubAsst
pctWRetire
medFamInc
perCapInc
whitePerCap
blackPerCap
indianPerCap
AsianPerCap
OtherPerCap
HispPerCap
NumUnderPov
PctPopUnderPov
PctLess9thGrade
PctNotHSGrad
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


2(c): Correlation matrix rows 32-62; all 122 predictor columns
PctBSorMore
PctUnemployed
PctEmploy
PctEmplManu
PctEmplProfServ
PctOccupManu
PctOccupMgmtProf
MalePctDivorce
MalePctNevMarr
FemalePctDiv
TotalPctDiv
PersPerFam
PctFam2Par
PctKids2Par
PctYoungKids2Par
PctTeen2Par
PctWorkMomYoungKids
PctWorkMom
NumIlleg
PctIlleg
NumImmig
PctImmigRecent
PctImmigRec5
PctImmigRec8
PctImmigRec10
PctRecentImmig
PctRecImmig5
PctRecImmig8
PctRecImmig10
PctSpeakEnglOnly
PctNotSpeakEnglWell
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


2(c): Correlation matrix rows 63-93; all 122 predictor columns
PctLargHouseFam
PctLargHouseOccup
PersPerOccupHous
PersPerOwnOccHous
PersPerRentOccHous
PctPersOwnOccup
PctPersDenseHous
PctHousLess3BR
MedNumBR
HousVacant
PctHousOccup
PctHousOwnOcc
PctVacantBoarded
PctVacMore6Mos
MedYrHousBuilt
PctHousNoPhone
PctWOFullPlumb
OwnOccLowQuart
OwnOccMedVal
OwnOccHiQuart
RentLowQ
RentMedian
RentHighQ
MedRent
MedRentPctHousInc
MedOwnCostPctInc
MedOwnCostPctIncNoMtg
NumInShelters
NumStreet
PctForeignBorn
PctBornSameState
population
householdsize
racepctblack
racePctWhite
racePctAsian
racePctHisp
agePct12t21
agePct12t29
agePct16t24
agePct65up
numbUrban
pctUrban
medIncome
pctWWage
pctWFarmSelf
pctWInvInc
pctWSocSec
pctWPubAsst
pctWRetire
medFamInc
perCapInc
whitePerCap
blackPerCap
indianPerCap
AsianPerCap
OtherPerCap
HispPerCap
NumUnderPov
PctPopUnderPov
PctLess9thGrade
PctNotHSGrad
PctBSorMore
PctUnemployed
PctEmploy
PctEmplManu
PctEmplProfServ
PctOccupManu
PctOccupMgmtProf
MalePctDivorce
MalePctNevMarr
FemalePctDiv
TotalPctDiv
PersPerFam
PctFam2Par
PctKids2Par
PctYoungKids2Par
PctTeen2Par
PctWorkMomYoungKids
PctWorkMom
NumIlleg
PctIlleg
NumImmig
PctImmigRecent
PctImmigRec5
PctImmigRec8
PctImmigRec10
PctRecentImmig
PctRecImmig5
PctRecImmig8
PctRecImmig10
PctSpeakEnglOnly
PctNotSpeakEnglWell
PctLargHouseFam
PctLargHouseOccup
PersPerOccupHous
PersPerOwnOccHous
PersPerRentOccHous
PctPersOwnOccup
PctPersDenseHous
PctHousLess3BR
MedNumBR
HousVacant
PctHousOccup
PctHousOwnOcc
PctVacantBoarded
PctVacMore6Mos
MedYrHousBuilt
PctHousNoPhone
PctWOFullPlumb
OwnOccLowQuart
OwnOccMedVal
OwnOccHiQuart
RentLowQ
RentMedian
RentHighQ
MedRent
MedRentPctHousInc
MedOwnCostPctInc
MedOwnCostPctIncNoMtg
NumInShelters
NumStreet
PctForeignBorn
PctBornSameState
PctSameHouse85
PctSameCity85
PctSameState85
LemasSwornFT
LemasSwFTPerPop
LemasSwFTFieldOps
LemasSwFTFieldPerPop
LemasTotalReq
LemasTotReqPerPop
PolicReqPerOffic
PolicPerPop
RacialMatchCommPol
PctPolicWhite
PctPolicBlack
PctPolicHisp
PctPolicAsian
PctPolicMinor
OfficAssgnDrugUnits
NumKindsDrugsSeiz
PolicAveOTWorked
LandArea
PopDens
PctUsePubTrans
PolicCars
PolicOperBudg
LemasPctPolicOnPatr
LemasGangUnitDeploy
LemasPctOfficDrugUn
PolicBudgPerPop
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


2(c): Correlation matrix rows 94-122; all 122 predictor columns
PctSameHouse85
PctSameCity85
PctSameState85
LemasSwornFT
LemasSwFTPerPop
LemasSwFTFieldOps
LemasSwFTFieldPerPop
LemasTotalReq
LemasTotReqPerPop
PolicReqPerOffic
PolicPerPop
RacialMatchCommPol
PctPolicWhite
PctPolicBlack
PctPolicHisp
PctPolicAsian
PctPolicMinor
OfficAssgnDrugUnits
NumKindsDrugsSeiz
PolicAveOTWorked
LandArea
PopDens
PctUsePubTrans
PolicCars
PolicOperBudg
LemasPctPolicOnPatr
LemasGangUnitDeploy
LemasPctOfficDrugUn
PolicBudgPerPop
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

2(d) Coefficient of variation for every predictor

CV = s/m uses sample standard deviation with ddof=1 and sample mean over the median-imputed training rows only. CV is computed before standardization. A nonzero constant with nonzero mean has CV=0; any zero (or numerically near-zero, absolute tolerance 1e-12) mean has undefined CV and is excluded from ranking. All-zero columns are also undefined. This rule avoids replacing an undefined ratio with an arbitrary large number. The target and identifier fields are excluded. CV is sensitive to the dataset's normalization origin and does not measure predictive importance.

Feature	Mean	Sample SD (ddof=1)	CV
PctPolicAsian	0.0162542	0.0898378	5.52706
NumStreet	0.0229833	0.0986654	4.29292
NumInShelters	0.0295987	0.102736	3.47095
NumIlleg	0.037204	0.113806	3.05896
NumImmig	0.0305953	0.0895413	2.92664
LemasPctOfficDrugUn	0.0933043	0.238201	2.55295
NumUnderPov	0.0568495	0.133167	2.34244
LemasSwornFT	0.0287492	0.0646293	2.24804
population	0.0581672	0.130359	2.2411
numbUrban	0.0646488	0.131784	2.03846
HousVacant	0.0764749	0.150538	1.96847
PolicOperBudg	0.0380736	0.0626504	1.64551
LandArea	0.064107	0.105482	1.64541
racePctHisp	0.144963	0.233694	1.61209
PctNotSpeakEnglWell	0.15188	0.221231	1.45662
LemasTotalReq	0.0497057	0.072167	1.45189
racepctblack	0.179839	0.25697	1.42889
PctUsePubTrans	0.165197	0.230799	1.39711
racePctAsian	0.153712	0.20891	1.3591
PctPolicHisp	0.063786	0.0863586	1.35388
PctRecentImmig	0.185077	0.238432	1.28829
PctRecImmig5	0.184943	0.237871	1.28618
PctRecImmig10	0.185318	0.235442	1.27048
PctRecImmig8	0.187893	0.237793	1.26558
OfficAssgnDrugUnits	0.0459331	0.0539514	1.17457
PctPersDenseHous	0.187398	0.212308	1.13293
PolicCars	0.0850234	0.0946167	1.11283
PctForeignBorn	0.218281	0.232661	1.06588
PctVacantBoarded	0.203318	0.214465	1.05483
PctHousNoPhone	0.265485	0.24498	0.922765

Feature	Mean	Sample SD (ddof=1)	CV
PctIlleg	0.249793	0.230196	0.92155
OwnOccMedVal	0.26297	0.233888	0.88941
OwnOccHiQuart	0.268722	0.237438	0.883583
PopDens	0.233759	0.202085	0.864499
OwnOccLowQuart	0.2641	0.227012	0.859566
PctWOFullPlumb	0.247552	0.208737	0.843206
indianPerCap	0.202555	0.165765	0.818368
MedNumBR	0.313043	0.254801	0.813947
PctPolicBlack	0.136776	0.106123	0.775887
PctLargHouseOccup	0.252261	0.192525	0.763198
PctPopUnderPov	0.30788	0.232286	0.75447
PctLargHouseFam	0.268555	0.198379	0.738692
pctWPubAsst	0.323445	0.228419	0.706206
pctWFarmSelf	0.290589	0.203964	0.701898
PctImmigRecent	0.324943	0.221444	0.681486
PctLess9thGrade	0.320147	0.217857	0.68049
OtherPerCap	0.281793	0.189725	0.673279
pctUrban	0.693318	0.446024	0.643318
RentLowQ	0.345773	0.221177	0.639662
AsianPerCap	0.326107	0.200512	0.614866
blackPerCap	0.29194	0.174971	0.599341
RentHighQ	0.421465	0.250375	0.594059
PctBSorMore	0.36204	0.214574	0.59268
medIncome	0.359217	0.212649	0.591978
PctImmigRec5	0.362896	0.214397	0.590793
RentMedian	0.371973	0.21157	0.568778
MedRent	0.383311	0.215974	0.563443
PctUnemployed	0.366602	0.206186	0.562425
perCapInc	0.350508	0.195612	0.558079
medFamInc	0.374883	0.202688	0.54067
PctNotHSGrad	0.387619	0.206893	0.533753
whitePerCap	0.368281	0.191443	0.519828
PctOccupManu	0.396094	0.204535	0.516379
PctEmplManu	0.40107	0.205961	0.51353
PctImmigRec8	0.401639	0.202857	0.505072

Feature	Mean	Sample SD (ddof=1)	CV
agePct16t24	0.335425	0.167816	0.50031
HispPerCap	0.387057	0.187362	0.484068
PersPerRentOccHous	0.401686	0.19026	0.473655
MedYrHousBuilt	0.489324	0.230355	0.470762
MedOwnCostPctIncNoMtg	0.405552	0.190693	0.470207
PctImmigRec10	0.42994	0.196804	0.457748
PctPolicMinor	0.210074	0.0953795	0.454029
PctVacMore6Mos	0.430328	0.188519	0.438082
PctOccupMgmtProf	0.440722	0.190687	0.43267
MedOwnCostPctInc	0.448308	0.186273	0.415502
agePct65up	0.428569	0.176903	0.412776
PolicBudgPerPop	0.156515	0.0639598	0.40865
MalePctNevMarr	0.435666	0.176036	0.404062
PctEmplProfServ	0.441512	0.177862	0.402849
MalePctDivorce	0.462622	0.184756	0.399366
LemasTotReqPerPop	0.17703	0.0684521	0.386669
TotalPctDiv	0.495478	0.185262	0.373905
agePct12t21	0.422649	0.155993	0.369083
PersPerOccupHous	0.45986	0.168885	0.367253
pctWInvInc	0.495786	0.181128	0.365335
pctWSocSec	0.475619	0.171563	0.360715
FemalePctDiv	0.488308	0.175981	0.36039
householdsize	0.461732	0.164284	0.3558
PctEmploy	0.495545	0.17416	0.35145
PctPersOwnOccup	0.561425	0.196055	0.34921
pctWRetire	0.478441	0.165264	0.345421
PolicPerPop	0.185572	0.0638862	0.344267
LemasSwFTPerPop	0.185572	0.0638862	0.344267
PctHousLess3BR	0.49794	0.171069	0.343553
MedRentPctHousInc	0.489819	0.16804	0.343065
PctWorkMomYoungKids	0.499492	0.170167	0.340681
PolicAveOTWorked	0.28697	0.0964996	0.336271
PctBornSameState	0.611311	0.20555	0.336245
PctHousOwnOcc	0.547017	0.183868	0.336129
PctWorkMom	0.524047	0.176089	0.336018

Feature	Mean	Sample SD (ddof=1)	CV
LemasGangUnitDeploy	0.492308	0.164974	0.335104
PctKids2Par	0.619538	0.207468	0.334876
PctSameHouse85	0.537365	0.179759	0.334519
PctYoungKids2Par	0.661157	0.220716	0.333833
PctFam2Par	0.609398	0.203048	0.333195
racePctWhite	0.752829	0.248594	0.330213
pctWWage	0.553498	0.181887	0.328614
PctTeen2Par	0.583264	0.191194	0.327801
PersPerOwnOccHous	0.49307	0.157395	0.319214
PctSameCity85	0.629572	0.200681	0.318758
PersPerFam	0.486682	0.154461	0.317375
PctSameState85	0.655906	0.195745	0.298434
agePct12t29	0.492087	0.143352	0.291315
PctSpeakEnglOnly	0.784047	0.227373	0.289999
LemasSwFTFieldPerPop	0.215639	0.0616798	0.286033
PctHousOccup	0.72103	0.193367	0.268182
PolicReqPerOffic	0.307124	0.0795142	0.2589
NumKindsDrugsSeiz	0.56713	0.0799774	0.141021
RacialMatchCommPol	0.722375	0.0936572	0.129652
PctPolicWhite	0.762856	0.090572	0.118727
LemasPctPolicOnPatr	0.752495	0.0843036	0.112032
LemasSwFTFieldOps	0.96188	0.0619583	0.0644138

Undefined CVs: **0**; constant training predictors: **0**. All 122 coefficients are reported above and in results/coefficient_of_variation.csv.

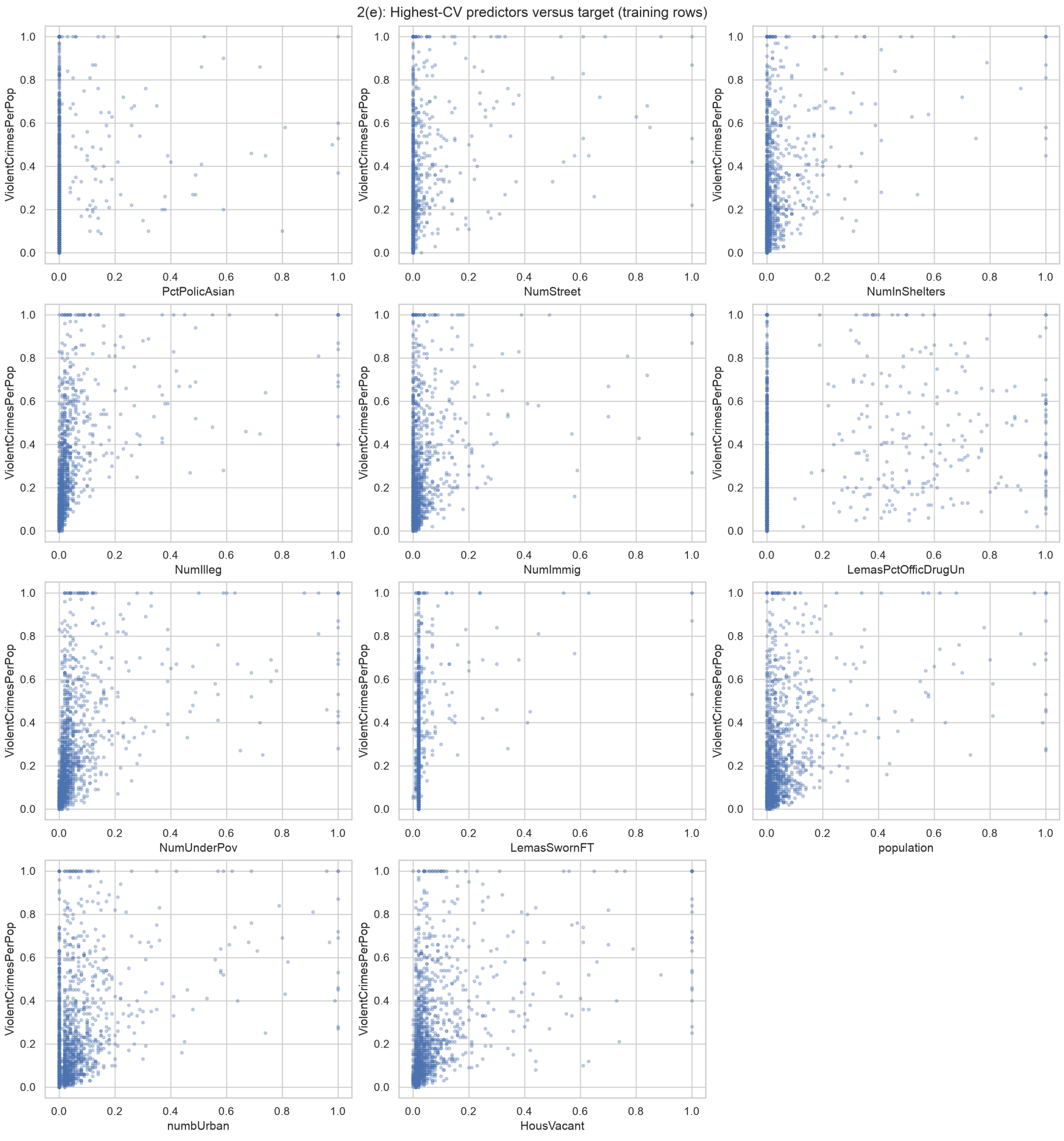
2(e) The 11 highest-CV predictors: scatter plots and box plots

The count is floor(sqrt(128))=11 as specified, even though only 122 columns are predictive. These plots use the same imputed training rows. Each scatter plot has a predictor on the horizontal axis and ViolentCrimesPerPop on the vertical axis; each box plot summarizes one predictor. The 11-feature selection does not restrict subsequent regression models.

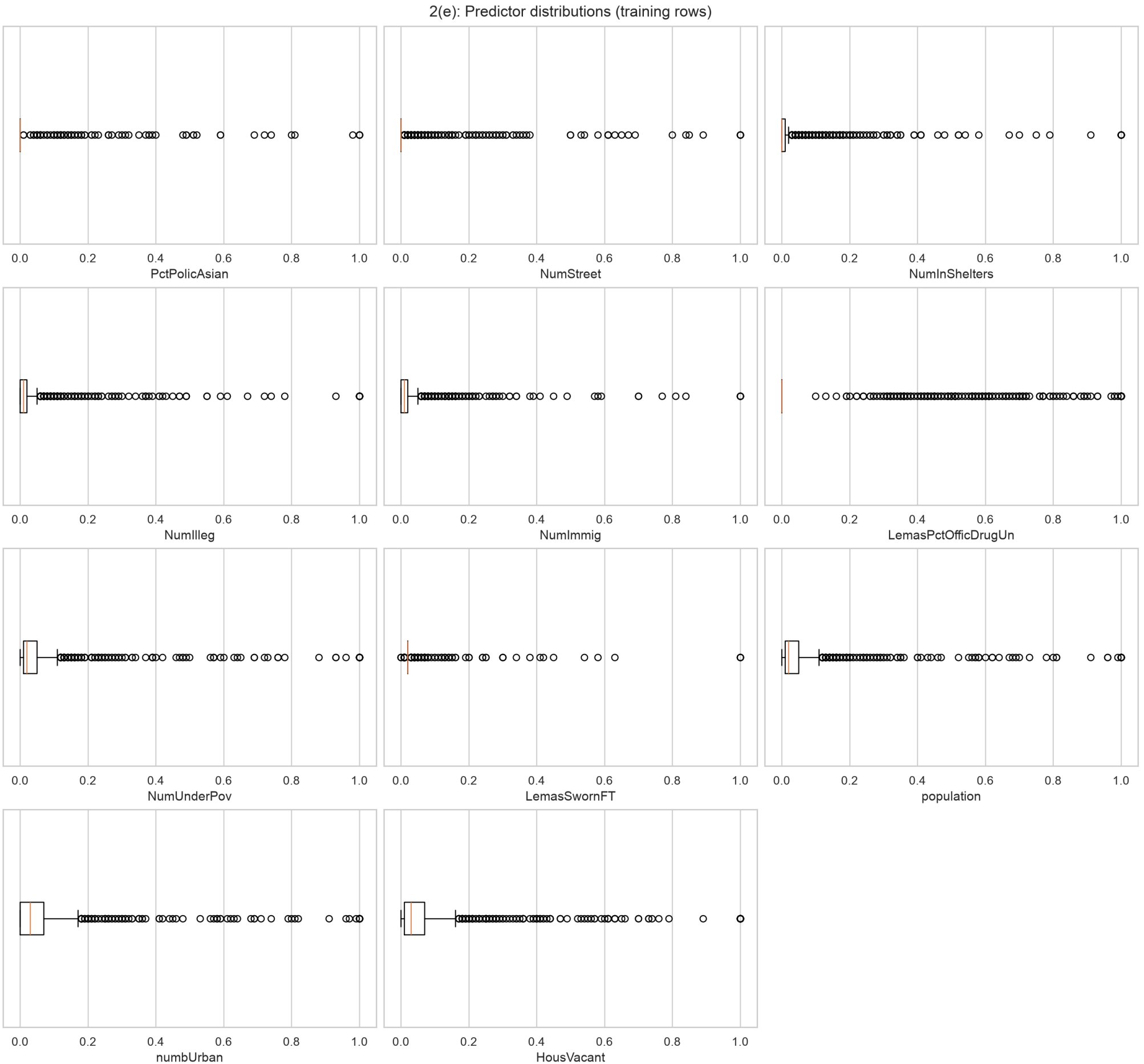
Scatter patterns can suggest skew, outliers, nonlinear associations, and heteroskedasticity. Neither scatter plots nor box plots establish statistical significance. Formal inference would require a specified null, a suitable model and assumptions, uncertainty estimates and multiple-comparison handling. Large CV alone does not imply a significant relationship with the target, and these observational data cannot establish causal effects.

Feature	Mean	Sample SD (ddof=1)	CV
PctPolicAsian	0.0162542	0.0898378	5.52706
NumStreet	0.0229833	0.0986654	4.29292
NumInShelters	0.0295987	0.102736	3.47095
NumIlleg	0.037204	0.113806	3.05896
NumImmig	0.0305953	0.0895413	2.92664
LemasPctOfficDrugUn	0.0933043	0.238201	2.55295
NumUnderPov	0.0568495	0.133167	2.34244
LemasSwornFT	0.0287492	0.0646293	2.24804
population	0.0581672	0.130359	2.2411
numbUrban	0.0646488	0.131784	2.03846
HousVacant	0.0764749	0.150538	1.96847

Scatter plots for the 11 highest training CV predictors; y is always the crime target.



Individual horizontal box plots of the selected predictors.



2(f) Ordinary least squares

OLS fits an intercept and all 122 imputed predictors by minimizing residual sum of squares. It has no tuned hyperparameter. Coefficients can be unstable under multicollinearity; the matrix rank and condition number below describe this concern. Main test error is MSE = mean((y-yhat)^2), with RMSE=sqrt(MSE) and R-squared also reported. All model test results are deliberately collected together after Question 2(j) has frozen every fitted model.

Centered training design rank = **121 / 122**; condition number = **3.51e+16**.

Final held-out result - OLS: MSE = 0.01796622; RMSE = 0.13403811; R-squared = 0.622204. This evaluation was performed after every model's training-only selection was complete.

2(g) Ridge regression

A Pipeline applies median imputation and StandardScaler before Ridge. Scikit-learn Ridge minimizes ||y-Xb||^2 + alpha*||b||^2, with an unpenalized intercept. Thus its alpha is the assignment's lambda under this convention. A 49-point logarithmic grid from 1e-6 through 1e6 is searched using training-fold MSE; if an artificial endpoint wins, the range is expanded. Standardizing equalizes penalty treatment across features, despite their existing [0,1] normalization.

Ridge: model__alpha = **56.23413251903491**; five-fold CV MSE = **0.01914446**; final grid [1e-06, 1000000.0], 49 candidates. Convergence warnings: **0**. The selected value is interior to the final search range.

Final held-out result - Ridge: MSE = 0.01747522; RMSE = 0.13219389; R-squared = 0.632528. This evaluation was performed after every model's training-only selection was complete.

2(h) LASSO without and with standardization

Both models use median imputation inside CV. The first retains the original supplied normalized units; the second applies fold-fitted StandardScaler. Lasso minimizes ||y-Xb||^2/(2*n) + alpha*||b||_1 with an unpenalized intercept, so numerical alpha is not directly comparable with Ridge alpha. Each model searches 41 logarithmically spaced values from 1e-5 to 1e-1, expanding artificial boundaries if needed. A strict tolerance of 1e-7 and max_iter=200000 are used; convergence warnings cause execution to fail instead of being hidden. Selected variables have absolute fitted coefficient greater than 1e-10. Coefficients for the standardized fit are expressed per training standard deviation; original-unit equivalents are also exported.

LASSO_unscaled: model__alpha = **0.0001**; five-fold CV MSE = **0.01899094**; final grid [1e-05, 0.1], 41 candidates. Convergence warnings: **0**. The selected value is interior to the final search range.

Feature	Fitted coefficient	Original-unit coefficient
racepctblack	0.268808	0.268808
racePctAsian	-0.0172314	-0.0172314
agePct12t21	0.074721	0.074721
agePct12t29	-0.291053	-0.291053
agePct65up	0.00200299	0.00200299
pctUrban	0.0412488	0.0412488
pctWWage	-0.0922778	-0.0922778
pctWFarmSelf	0.0290908	0.0290908
pctWInvInc	-0.11137	-0.11137
pctWSocSec	0.0607102	0.0607102
pctWPubAsst	0.0431714	0.0431714
pctWRetire	-0.109425	-0.109425
whitePerCap	-0.0635797	-0.0635797
blackPerCap	-0.0230025	-0.0230025
indianPerCap	-0.0313048	-0.0313048
AsianPerCap	0.0289075	0.0289075
OtherPerCap	0.0238765	0.0238765
HispPerCap	0.0246815	0.0246815
PctPopUnderPov	-0.14882	-0.14882
PctLess9thGrade	-0.0221712	-0.0221712
PctBSorMore	0.0139991	0.0139991
PctEmploy	0.155465	0.155465
PctEmplManu	-0.0185934	-0.0185934
PctOccupManu	0.0170117	0.0170117
MalePctDivorce	0.131065	0.131065
MalePctNevMarr	0.151274	0.151274
FemalePctDiv	-0.0981841	-0.0981841

Feature	Fitted coefficient	Original-unit coefficient
PctKids2Par	-0.200342	-0.200342
PctYoungKids2Par	-0.036761	-0.036761
PctTeen2Par	-0.0110275	-0.0110275
PctWorkMom	-0.134091	-0.134091
NumIlleg	-0.100974	-0.100974
PctIlleg	0.0970655	0.0970655
NumImmig	-0.105417	-0.105417
PctImmigRec10	-0.00481494	-0.00481494
PctRecImmig10	0.0408048	0.0408048
PctNotSpeakEnglWell	-0.0834929	-0.0834929
PctLargHouseOccup	-0.0871231	-0.0871231
PersPerOccupHous	0.0739089	0.0739089
PctPersOwnOccup	-0.0656861	-0.0656861
PctPersDenseHous	0.225854	0.225854
PctHousLess3BR	0.0664367	0.0664367
MedNumBR	0.00589584	0.00589584
HousVacant	0.128432	0.128432
PctHousOccup	-0.0607111	-0.0607111
PctVacantBoarded	0.0746789	0.0746789
PctVacMore6Mos	-0.0627848	-0.0627848
MedYrHousBuilt	-0.0152675	-0.0152675
PctHousNoPhone	0.0270152	0.0270152
PctWOFullPlumb	-0.0259594	-0.0259594
OwnOccLowQuart	-0.0202643	-0.0202643
RentLowQ	-0.165396	-0.165396
MedRent	0.173436	0.173436
MedRentPctHousInc	0.0906902	0.0906902
MedOwnCostPctInc	-0.0174525	-0.0174525
MedOwnCostPctIncNoMtg	-0.0798485	-0.0798485
NumInShelters	0.0695764	0.0695764
NumStreet	0.159943	0.159943
PctForeignBorn	0.0558053	0.0558053
PctSameCity85	0.0245015	0.0245015
LemasSwFTFieldPerPop	0.0754785	0.0754785
LemasTotalReq	-0.00747892	-0.00747892

Feature	Fitted coefficient	Original-unit coefficient
PolicReqPerOffic	0.0967116	0.0967116
RacialMatchCommPol	-0.0248653	-0.0248653
PctPolicBlack	-0.0190259	-0.0190259
PctPolicHisp	0.0231088	0.0231088
PctPolicAsian	0.0236631	0.0236631
NumKindsDrugsSeiz	-0.0353149	-0.0353149
PolicAveOTWorked	-0.0115196	-0.0115196
LandArea	0.0360547	0.0360547
PopDens	0.0133753	0.0133753
PctUsePubTrans	-0.0471896	-0.0471896
PolicCars	0.0307014	0.0307014
LemasPctPolicOnPatr	0.0160821	0.0160821
LemasGangUnitDeploy	0.0379637	0.0379637

LASSO_unscaled selects 75 variables: racepctblack, racePctAsian, agePct12t21, agePct12t29, agePct65up, pctUrban, pctWWage, pctWFarmSelf, pctWInvInc, pctWSocSec, pctWPubAsst, pctWRetire, whitePerCap, blackPerCap, indianPerCap, AsianPerCap, OtherPerCap, HispPerCap, PctPopUnderPov, PctLess9thGrade, PctBSorMore, PctEmploy, PctEmplManu, PctOccupManu, MalePctDivorce, MalePctNevMarr, FemalePctDiv, PctKids2Par, PctYoungKids2Par, PctTeen2Par, PctWorkMom, NumIlleg, PctIlleg, NumImmig, PctImmigRec10, PctRecImmig10, PctNotSpeakEnglWell, PctLargHouseOccup, PersPerOccupHous, PctPersOwnOccup, PctPersDenseHous, PctHousLess3BR, MedNumBR, HousVacant, PctHousOccup, PctVacantBoarded, PctVacMore6Mos, MedYrHousBuilt, PctHousNoPhone, PctWOFullPlumb, OwnOccLowQuart, RentLowQ, MedRent, MedRentPctHousInc, MedOwnCostPctInc, MedOwnCostPctIncNoMtg, NumInShelters, NumStreet, PctForeignBorn, PctSameCity85, LemasSwFTFieldPerPop, LemasTotalReq, PolicReqPerOffic, RacialMatchCommPol, PctPolicBlack, PctPolicHisp, PctPolicAsian, NumKindsDrugsSeiz, PolicAveOTWorked, LandArea, PopDens, PctUsePubTrans, PolicCars, LemasPctPolicOnPatr, LemasGangUnitDeploy. Final solver iterations: 573; dual gap: 5.54e-09.

LASSO_scaled: model__alpha = **0.000630957344480193**; five-fold CV MSE = **0.01910257**; final grid [1e-05, 0.1], 41 candidates. Convergence warnings: **0**. The selected value is interior to the final search range.

Feature	Fitted coefficient	Original-unit coefficient
racepctblack	0.0680053	0.264731
racePctAsian	-0.00186313	-0.00892129
agePct12t21	0.0124865	0.080072
agePct12t29	-0.043304	-0.302182
pctUrban	0.0171305	0.0384201
pctWWage	-0.0142389	-0.0783104
pctWFarmSelf	0.00484792	0.0237765
pctWInvInc	-0.0221439	-0.122296
pctWSocSec	0.0104043	0.0606643
pctWPubAsst	0.00791543	0.0346647
pctWRetire	-0.0178288	-0.107917

Feature	Fitted coefficient	Original-unit coefficient
whitePerCap	-0.00964295	-0.0503867
blackPerCap	-0.00390531	-0.0223272
indianPerCap	-0.00525503	-0.0317123
AsianPerCap	0.00544171	0.0271481
OtherPerCap	0.00407187	0.0214691
HispPerCap	0.00435151	0.0232329
PctPopUnderPov	-0.0293757	-0.126506
PctLess9thGrade	-0.00349112	-0.0160302
PctEmploy	0.0251218	0.144294
PctEmplManu	-0.00175552	-0.00852638
PctOccupManu	7.960e-04	0.00389285
MalePctDivorce	0.021307	0.115364
MalePctNevMarr	0.02474	0.140587
FemalePctDiv	-0.0146468	-0.0832571
PctKids2Par	-0.0406905	-0.196194
PctYoungKids2Par	-0.0077605	-0.0351723
PctTeen2Par	-0.00196093	-0.0102596
PctWorkMom	-0.0224057	-0.127283
NumIlleg	-0.0102618	-0.0901993
PctIlleg	0.0239366	0.104018
NumImmig	-0.0112994	-0.126234
PctImmigRecent	-1.529e-05	-6.909e-05
PctImmigRec10	-6.603e-04	-0.00335609
PctRecImmig10	0.00868903	0.0369176
PctNotSpeakEnglWell	-0.0131862	-0.0596236
PctLargHouseOccup	-0.0120755	-0.0627427
PersPerOccupHous	0.00926283	0.0548653
PersPerOwnOccHous	-3.780e-04	-0.00240222
PctPersOwnOccup	-0.0114699	-0.0585229
PctPersDenseHous	0.0433582	0.204291
PctHousLess3BR	0.011284	0.0659838
MedNumBR	7.359e-04	0.0028893
HousVacant	0.0175311	0.116495
PctHousOccup	-0.0118836	-0.0614768
PctVacantBoarded	0.0150717	0.0702995

Feature	Fitted coefficient	Original-unit coefficient
PctVacMore6Mos	-0.0110045	-0.0583928
MedYrHousBuilt	-0.00311572	-0.0135302
PctHousNoPhone	0.00359767	0.0146905
PctWOFullPlumb	-0.00513436	-0.0246055
OwnOccLowQuart	-0.00117517	-0.00517845
RentLowQ	-0.0308778	-0.139653
MedRent	0.0305419	0.141462
MedRentPctHousInc	0.0152763	0.0909392
MedOwnCostPctInc	-0.00363182	-0.0195039
MedOwnCostPctIncNoMtg	-0.0146295	-0.0767431
NumInShelters	0.00850313	0.0827949
NumStreet	0.0174653	0.177075
PctForeignBorn	0.00816733	0.0351157
PctSameCity85	0.00350471	0.0174699
LemasSwFTFieldPerPop	0.00713891	0.11578
LemasTotalReq	-0.00546536	-0.0757574
PolicReqPerOffic	0.00965535	0.12147
RacialMatchCommPol	-0.00317046	-0.0338631
PctPolicBlack	-0.0032841	-0.0309566
PctPolicHisp	0.00305382	0.0353739
PctPolicAsian	0.00208558	0.0232227
OfficAssgnDrugUnits	-0.00267528	-0.0496034
NumKindsDrugsSeiz	-0.00347365	-0.0434475
PolicAveOTWorked	-0.00193853	-0.0200953
LandArea	0.00522717	0.0495716
PopDens	0.00354127	0.0175296
PctUsePubTrans	-0.00977692	-0.0423754
PolicCars	0.00777032	0.0821517
LemasPctPolicOnPatr	0.00176573	0.0209519
LemasGangUnitDeploy	0.00629364	0.038162

LASSO_scaled selects 76 variables: racepctblack, racePctAsian, agePct12t21, agePct12t29, pctUrban, pctWWage, pctWFarmSelf, pctWInvInc, pctWSocSec, pctWPubAsst, pctWRetire, whitePerCap, blackPerCap, indianPerCap, AsianPerCap, OtherPerCap, HispPerCap, PctPopUnderPov, PctLess9thGrade, PctEmploy, PctEmplManu, PctOccupManu, MalePctDivorce, MalePctNevMarr, FemalePctDiv, PctKids2Par, PctYoungKids2Par, PctTeen2Par, PctWorkMom, NumIlleg, PctIlleg, NumImmig, PctImmigRecent, PctImmigRec10, PctRecImmig10, PctNotSpeakEnglWell, PctLargHouseOccup, PersPerOccupHous, PersPerOwnOccHous, PctPersOwnOccup, PctPersDenseHous, PctHousLess3BR, MedNumBR, HousVacant, PctHousOccup, PctVacantBoarded, PctVacMore6Mos, MedYrHousBuilt, PctHousNoPhone, PctWOFullPlumb, OwnOccLowQuart, RentLowQ,

MedRent, MedRentPctHousInc, MedOwnCostPctInc, MedOwnCostPctIncNoMtg, NumInShelters, NumStreet, PctForeignBorn, PctSameCity85, LemasSwFTFieldPerPop, LemasTotalReq, PolicReqPerOffic, RacialMatchCommPol, PctPolicBlack, PctPolicHisp, PctPolicAsian, OfficAssgnDrugUnits, NumKindsDrugsSeiz, PolicAveOTWorked, LandArea, PopDens, PctUsePubTrans, PolicCars, LemasPctPolicOnPatr, LemasGangUnitDeploy. Final solver iterations: 546; dual gap: 4.74e-09.

Standardization changes the effective L1 penalty in original units: a unit change in a low-variance feature is treated differently after scaling. Hence the fitted supports and optimal penalties can differ. Correlated predictors may substitute for each other, so an excluded variable is not necessarily irrelevant. The test-error comparison follows in the frozen evaluation table.

Final held-out result - LASSO_unscaled: MSE = 0.01765779; RMSE = 0.13288261; R-squared = 0.628689. This evaluation was performed after every model's training-only selection was complete.

Final held-out result - LASSO_scaled: MSE = 0.01772176; RMSE = 0.13312311; R-squared = 0.627344. This evaluation was performed after every model's training-only selection was complete.

2(i) Principal component regression (PCR)

The Pipeline is median imputation -> standardization -> full-SVD PCA -> ordinary least squares. M is selected by the same five training folds and validation MSE. All integer values 1 through 122 are searched, so neither endpoint is an artificial truncated search boundary. PCA directions, imputation medians and scaling statistics are refitted within each fold. PCA maximizes predictor variance, not association with the response; a high explained-variance fraction alone does not determine the best M.

PCR: pca__n_components = **93**; five-fold CV MSE = **0.01944865**; final grid [1, 122], 122 candidates. Convergence warnings: **0**. The selected value is interior to the final search range.

The selected 93 components explain **99.7959%** of standardized training predictor variance.

Final held-out result - PCR: MSE = 0.01826425; RMSE = 0.13514528; R-squared = 0.615937. This evaluation was performed after every model's training-only selection was complete.

2(j) L1-penalized boosting with XGBoost

Implementation assumption: interpret the allowed XGBoost option as **booster="gbtree"** with a CV-selected **reg_alpha**, an L1 penalty on scalar leaf weights. Standard tree leaves predict constants; they do not fit LASSO regressions at nodes. This differs from a literal tree with local multivariate L1 regressions. The standard gbtree implementation fulfills the explicit XGBoost option under this stated interpretation; no unsupported local-linear-tree claim is made. gblinear would fit a linear booster and would not be a boosting tree.

For a fixed tree structure, the regularizer includes reg_alpha*sum(|w_j|) + (reg_lambda/2)*sum(w_j^2). The L1 term can shrink leaf weights to zero and affects split gains. It is not direct coefficient selection over the original 122 features. See the official XGBoost parameter documentation and boosted-tree introduction.

The fixed, predeclared configuration uses squared-error loss, 400 trees, learning_rate=0.04, max_depth=3, min_child_weight=5, reg_lambda=1, subsample=1, colsample_bytree=1, gamma=0, hist trees, n_jobs=1 and seed 552. Only reg_alpha is tuned: zero plus 29 positive log-spaced candidates from 1e-5 through 1e2, expanding if necessary. Median imputation remains inside the Pipeline for consistent data handling. No test-set early stopping is used; other tree settings are fixed modeling assumptions, not claimed globally optimal.

XGBoost: model__reg_alpha = **0.00017782794100389227**; five-fold CV MSE = **0.01970332**; final grid [0.0, 100.0], 30 candidates. Convergence warnings: **0**. The selected value is interior to the final search range.

Model	Parameter	Selected	CV MSE	Grid min	Grid max	Candidates	Solver warnings
Ridge	model__alpha	56.2341	0.0191445	1.000e-06	1e+06	49	0
LASSO_unscaled	model__alpha	1.000e-04	0.0189909	1.000e-05	0.1	41	0
LASSO_scaled	model__alpha	6.310e-04	0.0191026	1.000e-05	0.1	41	0
PCR	pca__n_components	93	0.0194487	1	122	122	0
XGBoost	model__reg_alpha	1.778e-04	0.0197033	0	100	30	0

Model	Selected parameter	CV MSE	Test MSE	Test RMSE	Test R2	Selected variables
OLS	{"method": "least squares"}	N/A	0.0179662	0.134038	0.622204	N/A
Ridge	{"model__alpha": 56.23413251903491}	0.0191445	0.0174752	0.132194	0.632528	N/A
LASSO_unscaled	{"model__alpha": 0.0001}	0.0189909	0.0176578	0.132883	0.628689	75
LASSO_scaled	{"model__alpha": 0.000630957344480193}	0.0191026	0.0177218	0.133123	0.627344	76
PCR	{"pca__n_components": 93}	0.0194487	0.0182642	0.135145	0.615937	N/A
XGBoost	{"model__reg_alpha": 0.00017782794100389227}	0.0197033	0.0162141	0.127334	0.659048	N/A

OLS: test MSE = 0.01796622; RMSE = 0.13403811; R-squared = 0.622204.

Ridge: test MSE = 0.01747522; RMSE = 0.13219389; R-squared = 0.632528.

LASSO_unscaled: test MSE = 0.01765779; RMSE = 0.13288261; R-squared = 0.628689.

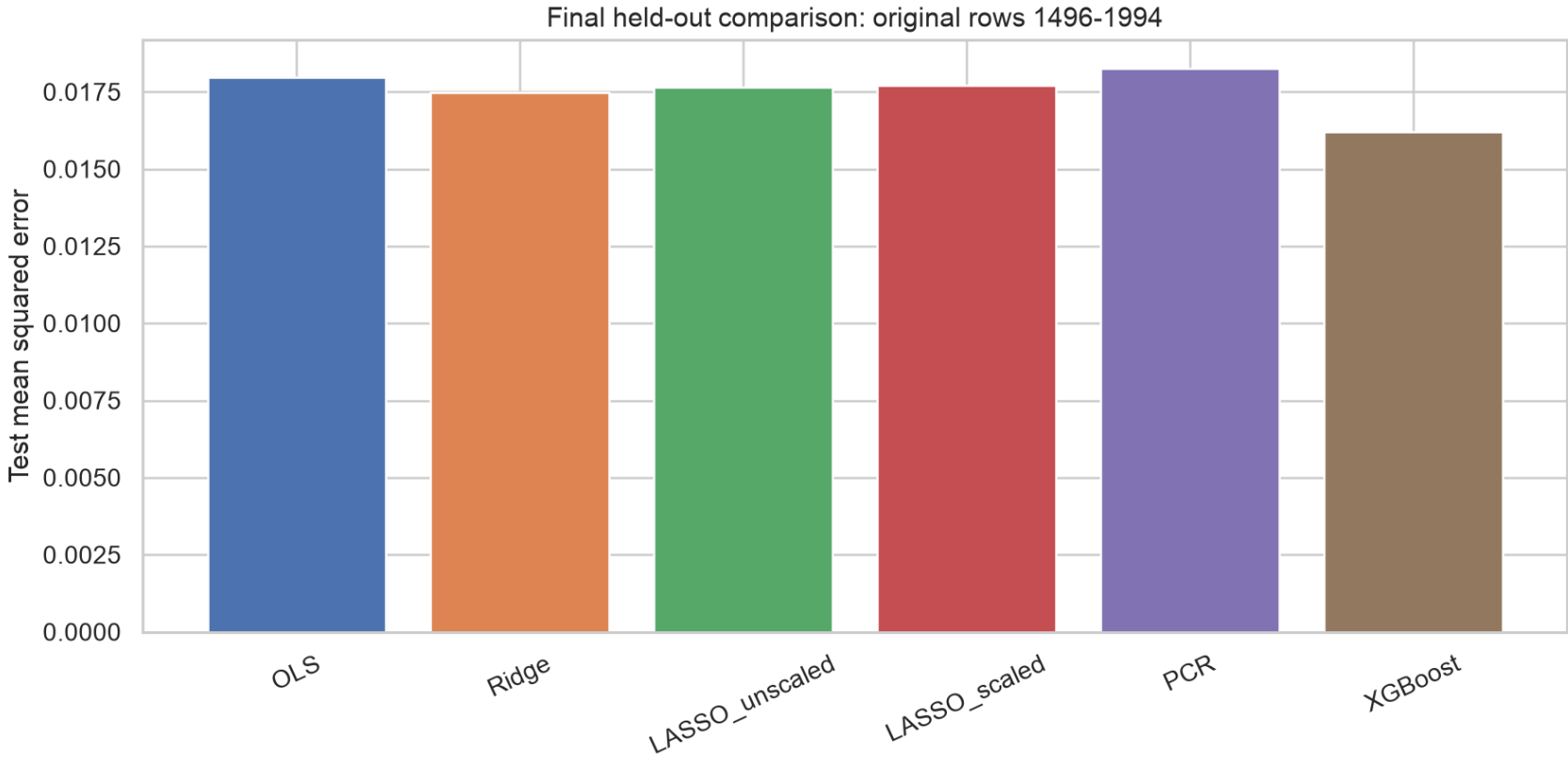
LASSO_scaled: test MSE = 0.01772176; RMSE = 0.13312311; R-squared = 0.627344.

PCR: test MSE = 0.01826425; RMSE = 0.13514528; R-squared = 0.615937.

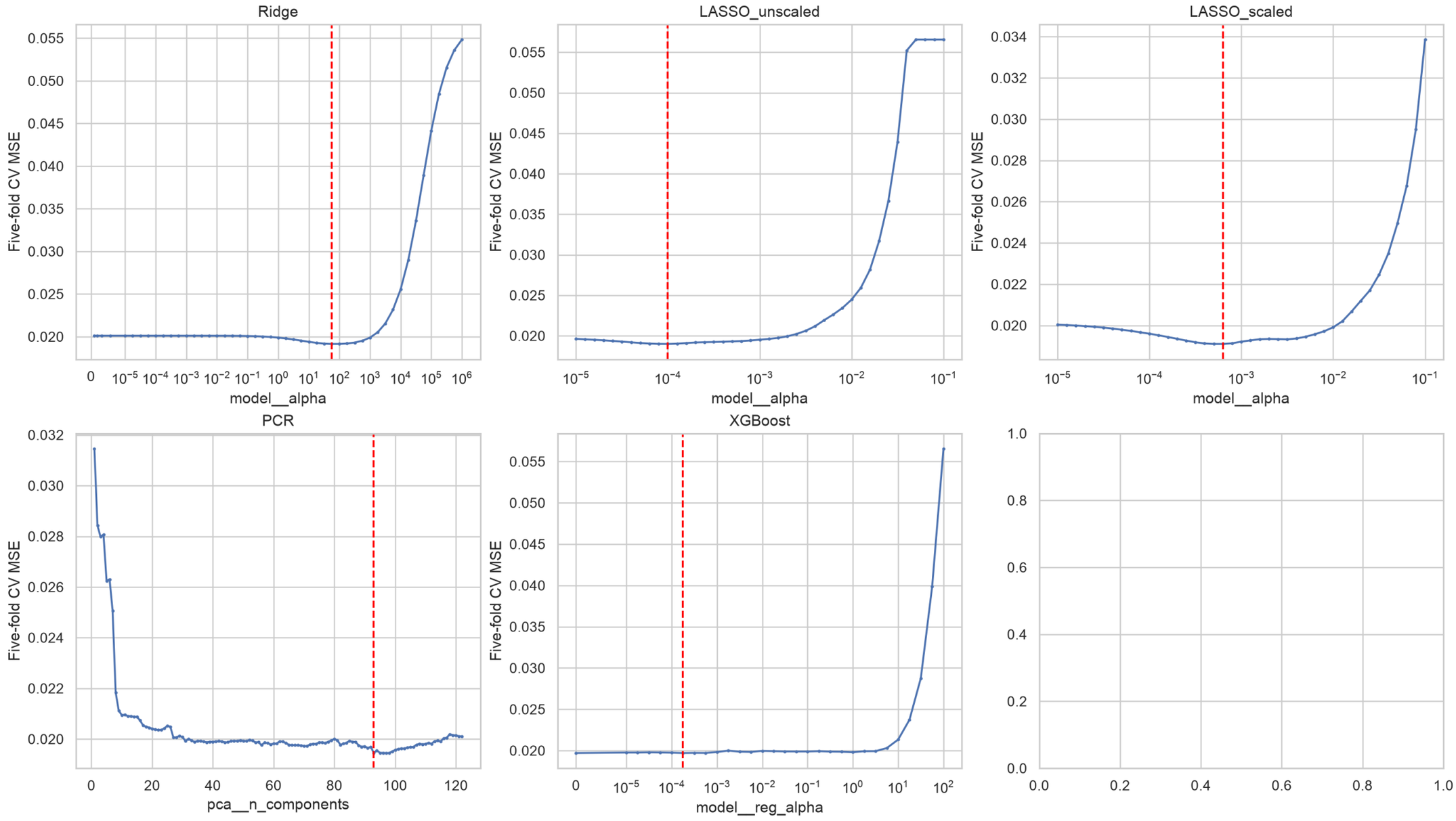
XGBoost: test MSE = 0.01621407; RMSE = 0.12733449; R-squared = 0.659048.

Standardized LASSO selected **76** variables versus **75** without standardization. Its test MSE is **0.00006398 higher**. This is a final held-out comparison, not a reason to retune. Lowest observed test MSE among the prespecified fits: **XGBoost**. This descriptive ranking on one fixed split does not establish a statistically reliable winner or estimate the performance of a newly selected model.

One final test evaluation for each frozen model. No test-based tuning.



Training-only hyperparameter selection curves. Dashed lines mark selected settings.



Reproducibility, validation and limitations

The notebook must run from the project root. All input and output paths are relative. Pinned dependencies, raw data hashes, fold assignments, full search results, LASSO coefficient tables, rules and test predictions are included. Every code cell is executed in order by a fresh Jupyter kernel using scripts/run_notebook.py. scripts/make_report.py creates the English PDF from the same recorded narrative, tables and figures, so the report is tied to computed outputs.

The original nonrandom external split may have distribution shifts; shuffled internal CV assumes exchangeability within the training portion. It does not account for possible geographic or social dependence. Median filling of heavily missing police variables is an information limitation, not a claim that missingness is ignorable. The 11 highest-CV features are not a formal inferential screen. Model rankings are specific to this split and stated configurations. Hyperparameter searches resolve grid endpoints, but do not guarantee a continuous global optimum.

No additional course submission naming, upload, repository, or packaging rules appear in the supplied homework PDF. Only the supplied instructions are asserted here.

Item	Verified requirement	Status
1(a)	120 rows, six features, two diagnoses; local data parsed	Complete
1(b)	Full-data fitted tree, complete plot and label mapping	Complete
1(c)	Every full-tree root-to-leaf rule; exact prediction agreement	Complete
1(d)	Cost-complexity CV, complexity/performance comparison, selected tree and all rules	Complete
2(a)	1994 x 128; original first 1495/last 499 split	Complete
2(b)	Five identifiers removed; fold-local median imputation; 122 predictors retained	Complete
2(c)	Full predictor correlation matrix plus readable panels	Complete
2(d)	All predictor CVs; ddof=1; training-only scope and edge cases	Complete
2(e)	Exactly 11 features; scatter and box plots; significance answer	Complete
2(f)	OLS fitted; test MSE, RMSE, R2	Complete
2(g)	Ridge CV alpha; test metrics	Complete
2(h)	Both LASSOs; CV alpha, support names/counts, convergence, comparison	Complete
2(i)	PCA plus OLS; all M searched by fold-local CV; test metrics	Complete
2(j)	gbtree/reg_alpha CV; explicit interpretation; test metrics	Complete

Outstanding assignment items: none. A literal local-LASSO model tree is not claimed; the documented standard-XGBoost interpretation of 2(j) is used. Unprovided course submission rules cannot be verified. Final PDF layout and fresh-kernel execution are additionally checked by the packaging workflow.

References

- Supplied Homework5.pdf, Questions 1-2 and footnotes 1-5.

- Supplied diagnosis.names and diagnosis.data. J. Czerniak and H. Zarzycki, Application of rough sets in the presumptive diagnosis of urinary system diseases, ACS 2002, 2003, pp. 41-51.

- Supplied communities.names and communities.data, UCI Machine Learning Repository; source data: US Census 1990, LEMAS 1990, FBI Crime in the United States 1995. M. A. Redmond and A. Baveja, A Data-Driven Software Tool for Enabling Cooperative Information Sharing Among Police Departments, European Journal of Operational Research 141 (2002), 660-678.

- Matthew Mayo, Simplifying Decision Tree Interpretability with Python & Scikit-learn, KDnuggets (2017), supplied PDF. The path traversal here is a fresh implementation with label decoding and prediction assertions.

- Scikit-learn Ridge documentation and Lasso documentation, for objective conventions.

- XGBoost parameters and boosted-tree introduction, for booster and leaf-weight regularization semantics.